The Ungoverned Substrate: Network States, Blockchain Governance, and the Option Both Programs Presuppose Away

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Abstract

Two contemporary research programs have addressed the question of what kind of polity a network-native coordination architecture might support. The first, articulated most prominently by Srinivasan [2022], proposes the *network state*: a community that forms online, accumulates social cohesion and shared institutions, and ultimately seeks territorial and diplomatic recognition as a sovereign-equivalent entity. The second, developed by De Filippi and Wright [2018] and the collaborators around the Coalition of Automated Legal Applications (COALA), examines the legal and governance consequences of running rules through code: *lex cryptographica*, decentralized autonomous organizations, on-chain governance, and the legal scaffolding such arrangements require to operate alongside conventional jurisdictions. The two programs differ in goal — one constitutional, the other infrastructural — but share a structural premise.

The shared premise is that the alternative-to-state-coordination must itself be a *governed entity*. The network state governs through a constitutional process inherited from the modern sovereign-state form. The blockchain-governance program governs through on-chain voting, off-chain coordination, and hybrid legal-computational arrangements. Both programs treat governance as a question that arises at the network or protocol layer and must be answered there. Both programs accordingly spend substantial effort on the design and analysis of governance mechanisms appropriate to their respective forms.

This paper observes that the bonded settlement primitive examined in the corpus's companion mechanism, institutional, and political-economy papers admits a third option that neither program names: an *ungoverned substrate* on which arbitrary governed entities can be constituted, but which itself is not a governed entity at all. The substrate does not vote; it does not upgrade; it does not have a constitution; it has no representatives, no executive, no judiciary, no protocol-level discretion of any kind. It executes a small set of invariant rules and admits arbitrary higher-layer arrangements to be composed on top of it. Network states can be constituted on the substrate; so can decentralized autonomous organizations; so can cooperatives, peer networks, individual sole traders, ad-hoc agent coalitions, and patterns the substrate's authors cannot anticipate. The substrate's freedom from governance is what enables the diversity of governance forms above it.

The substantive claim of this paper is therefore not that network-state or blockchain-governance programs are wrong on their own terms —- both are coherent research programs addressing real questions —- but that the architectural option they presuppose away is the option the bonded primitive realizes. Recognizing this option does not adjudicate among governed arrangements; it relocates the governance question from the substrate, where both programs place it, to the graph, where the substrate admits it to be answered in arbitrarily many ways. That relocation is the political-architectural content of the substrate change with respect to network polity, and it is the subject of this paper.

Keywords: network state, blockchain governance, lex cryptographica, ungoverned substrate, decentralized autonomous organization, infrastructural literacy, governance at the graph

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1 Introduction

This paper sits between political philosophy, the contemporary literature on blockchain governance, and the analysis of a particular settlement primitive. Its starting point is a question that the companion political-philosophy paper on coercion left open: when the state's sovereign-coercive function is partially substituted by bonded pre-commitment in bilateral commerce, what follows for the proposals that have emerged to organize coordination at the network layer? The question is sharp enough to deserve dedicated treatment.

Two such proposals have organized the recent discussion. The first is the network-state thesis, advanced most prominently by Srinivasan [2022] and adjacent commentators. The second is the blockchain-governance research program, developed by De Filippi and Wright [2018] and collaborators around COALA and parallel academic and practitioner communities. The two are distinct: the first is normative-prescriptive (a program for constituting alternative polities), the second is descriptive-analytical (an account of what blockchain arrangements imply for law and governance). Both, however, share a structural assumption that the present paper identifies and argues is not forced.

The shared assumption is that the network-layer alternative to state coordination must itself be a *governed entity*. The network state requires a constitutional founder, a community self-conception, an explicit governance process, and ultimately sovereign-equivalent diplomatic standing (Srinivasan 2022). The blockchain-governance program treats the governance of on-chain arrangements as the central problem of the field: how decisions are made about the protocol's evolution, who has voting power, what counts as legitimate amendment, how governance failures are detected and corrected (De Filippi et al. 2022). In both programs, the substrate of network-native coordination is itself something that must be governed, and the design and analysis of that governance is the central work.

The bonded settlement primitive examined in the corpus's companion mechanism paper provides a different option that neither program names: an *ungoverned substrate*. The kernel is invariant. It admits no upgrades, no proposals, no votes, no protocol-level discretion. There is no admin, no owner, no fee parameter to vote on, no governance token to hold. There is no entity of which the kernel is the constitution, because the kernel is not a constitution of anything. It is a small set of rules executed on a public infrastructure, on top of which arbitrary governed entities (network states, decentralized autonomous organizations, cooperatives, peer networks, sole traders, agent coalitions) can be composed — or which bilateral parties can use without constituting any entity at all.

The thesis. The thesis is structural. (1) Both the network-state program and the blockchain-governance program assume that the alternative to state coordination must be a governed network entity. (2) The assumption is not architecturally forced: a coordination substrate can be invariant rather than governed, and the decision to make it one or the other is a design choice. (3) The ungoverned-substrate option, where it is taken, does not abolish governance — it relocates governance from the substrate layer to the graph layer, where any pattern of governance (or absence of governance) can be composed. (4) The relocation is the political-architectural content of the substrate change, and it clarifies what each program is actually proposing: the network state is one possible graph, blockchain-governance arrangements are another, and neither is the substrate.

What this paper is and is not. This paper is a short essay in the political architecture of network coordination. It does not extend the underlying mechanism, the institutional argument, or the political-economy diagnostic of the companion papers, nor does it extend the political-philosophy of coercion treated in the companion paper on the coercion variable. It engages two contemporary research programs to clarify what the substrate change implies and does not imply for the proposals those programs have advanced.

The paper takes no position on whether network states are desirable, whether DAOs are well-governed, whether on-chain voting is preferable to off-chain coordination, whether *lex cryptographica* should be recognized by conventional jurisdictions, whether COALA's model legal frameworks are correctly drafted, or whether any other normative or prescriptive question of network polity has the right answer. Those are real questions, and the paper's silence on them is intentional: the substrate change is a precondition for those questions to be posed coherently at the graph layer, not an adjudication of how they should be answered.

Position relative to the corpus. This paper extends the political-philosophy treatment of the companion paper on coercion. That paper identified coercion as a substrate variable: structurally load-bearing in the dominant tradition's account of bilateral commercial enforcement, substituted by pre-commitment when the bonded primitive is the operable architecture. The present paper identifies governance as a related substrate variable: structurally load-bearing in both contemporary network-polity programs, substituted by *nothing* (or, equivalently, by an invariant kernel) when the bonded primitive is the operable architecture, and relocated to the graph for arrangements that require it. The two papers together complete the substrate-relocation argument for the political-philosophy cluster, as the institutional and political-economy companions did for the firm and the left/right partition.

2 The Network-State Program and Its Coordination Gap

We begin with the network-state thesis, both because it is the more public of the two programs and because the companion political-economy paper on hegemony identified its specific unsolved sub-problem as one the bonded primitive resolves.

The thesis as articulated. Srinivasan [2022] proposes the network state as a sequence: an online community forms around a shared interest, acquires social cohesion through ongoing interaction, develops shared institutions (norms, currencies, dispute mechanisms), attains a population large enough to matter and a treasury large enough to act, acquires territorial presence (initially distributed, eventually contiguous), and ultimately seeks diplomatic recognition as a sovereign-equivalent entity. The sequence is presented as a contemporary updating of Hirschman's framework on *exit* as a response to institutional dissatisfaction (Hirschman 1970): when voice within incumbent states becomes ineffective, exit to a new entity becomes the alternative, but exit requires a destination, and the network state is the destination's design.

The thesis takes governance as constitutive. A network state is not a network state without explicit shared institutions, without a self-conception articulated by a founder or coalition, without a process for collective decision, without internal adjudication of disputes, and ultimately without external recognition by other sovereigns. Each of these is a governance function, and the design of each is part of what the network-state program addresses.

The unsolved sub-problem. The thesis names an "economic engine" as a constitutive stage between community formation and recognition — the network state must be capable of supporting economic activity within its participants without routing the activity through incumbent state-clearable infrastructure — but treats that stage as relatively under-specified. In practice, commerce within network-state communities continues to route through fiat rails, incumbent platforms, and the legal apparatuses of the participants' physical jurisdictions. The companion political-economy paper on hegemony noted that this is not a minor implementation detail: the missing economic engine is the substrate gap, and the bonded primitive supplies it.

The supply is structural. Bilateral commercial coordination between network-state participants, between participants and non-participants, and between two non-participants who happen to share the substrate, can be conducted through bonded commitments that depend on no incumbent state apparatus and no network-state-internal governance. The primitive does not constitute the network state; it is the substrate on which network states (and other arrangements) can be constituted without retaining their dependency on the very state apparatuses they are exiting from.

What the thesis does not require. A subtler observation is that the substrate the network-state program needs does not have to be one the network state *governs*. Treating the substrate as something the network state should constitute under its own governance — a network-state-native blockchain, a network-state-issued currency governed by the network state's process, a network-state-operated arbitration system — recreates within the network state the very substrate-governance problem the network state was supposed to resolve relative to the incumbent state. The substrate the network state actually requires is one that no entity governs, which any entity (the network state, the incumbent state, a private cooperative, a single trader) can use, and whose neutrality is what makes it suitable for cross-affiliation coordination.

This is the option the network-state program presupposes away when it treats every functional layer as something the network state must eventually govern. The bonded primitive's substrate is not a layer of the network state; it is a layer beneath the network state and beneath the incumbent state alike, on which both can compose, and which is not the property of either.

Remark 2.1 (On exit and the missing destination). Hirschman's analysis of exit assumes that exit is meaningful only when it has a destination. The network-state program is constructive about destinations: it specifies what the destination should look like (a sovereign-equivalent network entity) and what work is required to constitute it. The bonded primitive contributes to the analysis of exit a different observation: exit can be partial. A participant can exit the incumbent state's commercial-coordination apparatus (by using the bonded substrate for commerce) without exiting the incumbent state's other functions (citizenship, criminal-law protection, public goods, defense). Partial exit was difficult under prior architectures because the incumbent state's commercial-coordination apparatus was bundled with its other functions. The substrate change unbundles. Whether the participant goes on to constitute a network state is a separate choice that the substrate does not make for them.

3 Blockchain Governance and the Confidence Machine

We turn now to the blockchain-governance research program, which addresses the same substrate question from a different angle and arrives at a more textured account of what governing network-native coordination involves.

The program in outline. De Filippi and Wright [2018] examine the legal consequences of running rules through code: *lex cryptographica* as an emerging body of computational law, smart contracts as self-executing agreements, decentralized autonomous organizations as code-mediated coordination structures, and the hybrid arrangements that emerge when computational rules encounter conventional jurisdictions. The framing extends the "code is law" formulation of Lessig [2006] into a positive program: where Lessig observed that code increasingly performs the regulatory function law had historically performed, the De Filippi–Wright account asks what it would mean for code to become recognized law in its own right, with its own forms, its own jurisprudence, and its own integration with conventional legal systems. The work has been extended by parallel research on on-chain governance design, the legal recognition of DAOs in specific jurisdictions, model legal frameworks (developed by COALA and adopted in modified form by several U.S. state legislatures), and the relationship between code-based arrangements and conventional legal institutions.

A consistent thread across this work is that the apparent elimination of governance through code is illusory. Rules are written by people; rules require interpretation when situations arise that the rules' authors did not anticipate; rule changes are required when external conditions shift; and the infrastructure on which the rules run (validators, sequencers, exchanges, fiat on-ramps) introduces governance through the back door even when the on-chain protocol claims to have eliminated it. The blockchain-governance program's central observation is that code-mediated coordination does not abolish governance — it relocates and transforms governance, often in ways that are less visible and accountable than the conventional institutions governance has historically inhabited.

The confidence machine. De Filippi et al. [2022], building on adjacent work by Reijers and Coeckelbergh [2018] that read blockchain as a narrative-technology shaping its participants' conceptions of trust and value, sharpen this observation into the framing of blockchain as a *confidence machine*: an architecture that shifts the locus of trust from people and institutions to code and mathematics, but that does not eliminate trust or governance. Confidence in code is required; the code is written and maintained by people; the people are governed by their own arrangements (corporate, foundational, informal); the infrastructure on which the code runs introduces further dependencies; and the participants' *infrastructural literacy* —- their capacity to evaluate the arrangements they are participating in — is generally inadequate to the arrangements' complexity.

The confidence-machine framing is important and largely correct for the class of arrangements it addresses, which is the class of *governance-rich* blockchain protocols: systems whose protocol-level rules are intended to evolve over time, whose parameters are subject to vote, whose treasuries fund ongoing development, and whose legitimacy depends on participants' continued endorsement of the governance process. For such systems, the confidence-machine analysis exposes the governance-of-the-governance problem clearly: trust shifted from human institutions to code does not arrive at a final resting place; it shifts to the human institutions that govern the code.

The COALA legal-scaffolding program. A complementary thread of work, also developed substantially within COALA, has produced model legal frameworks for DAOs, templates for the integration of on-chain arrangements with conventional legal personality, and analyses of how code-mediated arrangements can be made compatible with the expectations of courts, regulators, and counterparties operating under conventional legal systems (De Filippi and Hassan 2016). The work is collaborative rather than antagonistic: it treats code-mediated arrangements and conventional legal arrangements as complementary rather than alternatives, and seeks to provide the institutional scaffolding under which the two can coexist.

This is rigorous and useful work. It identifies that DAOs constituted as code-mediated entities still need legal recognition to enter into contracts with non-DAO counterparties, to hold property in conventional jurisdictions, to limit liability for participants, and to engage with regulators on terms regulators can recognize. The COALA model laws are attempts to make this scaffolding available in standardized form, so that DAOs need not each negotiate their legal status ad hoc.

Remark 3.1 (On the scope of the confidence-machine analysis). The confidence-machine analysis applies to the class of arrangements for which it was developed: governance-rich blockchain protocols whose parameters and rules evolve under explicit governance processes. It does not, on its own terms, require that all blockchain arrangements be of this kind. The analysis is silent on what happens when an architecture is *not* governance-rich — when its rules are invariant, when no parameter is subject to vote, and when no treasury funds the protocol's evolution. The present paper does not contradict the confidence-machine analysis; it identifies a class of arrangements outside its analytical scope.

4 The Common Premise: Governance at the Substrate Layer

We now identify the structural premise that the network-state program and the blockchain-governance program share, and that the present paper argues is not architecturally forced.

What both programs assume. The network-state program assumes that the alternative to state coordination is a network polity that must be governed: it has a founder, a constitution, a process for collective decision, a treasury managed by some authority, internal adjudication, and external diplomatic standing.

The blockchain-governance program assumes that the alternative to platform-mediated coordination is an on-chain protocol whose rules and parameters must be governed: it has a governance token, a voting process, a developer foundation, a process for upgrades, and external legal scaffolding to integrate with conventional jurisdictions.

In both cases, the alternative arrangement is itself a governed entity. The shape of the governance differs — constitutional in the network-state program, computational-democratic in the blockchain-governance program — but the structural location of the governance function is the same: at the network or protocol layer, where decisions about the system's rules, parameters, and evolution are made. Both programs accordingly devote substantial effort to the design and analysis of governance mechanisms appropriate to their respective forms.

Why the assumption seems necessary. The assumption is not arbitrary. Most blockchain protocols of which the contemporary research community is aware are governance-rich: their parameters change over time, their rules admit upgrade paths, their treasuries fund ongoing development. The largest and most-studied protocols (Ethereum, Bitcoin through its informal but real social-coordination process, Tezos, Cosmos chains, Optimism, Maker) all have governance processes of varying degrees of formality. The blockchain-governance program emerged as a rigorous account of these processes, and the network-state program emerged in parallel to address the related question of what such processes imply for the political organization of the communities they govern. The assumption that network-native coordination requires governance reflects the architectures the programs were analyzing.

Why the assumption is not architecturally forced. A protocol's parameters and rules need not evolve. A coordination architecture can be designed such that its small invariant kernel is fixed at deployment, with all variability pushed to layers above the kernel that the kernel does not constrain. The bonded settlement primitive examined in the corpus's companion mechanism paper is such an architecture. The kernel has no admin, no owner, no upgrade path, no governance token, no treasury, no parameter that can be voted on, no discretion of any kind. It executes a small fixed set of rules. Whatever assemblies are constituted on top of it inherit no governance dependency on the kernel and no obligation to participate in the kernel's evolution because the kernel does not evolve.

This is not an attempt to evade governance; it is a demonstration that governance is a substrate-design choice, not a substrate-design necessity. The choice has consequences. A governance-rich protocol can adapt to changing conditions; an invariant kernel cannot. A governance-rich protocol can correct errors in its rules; an invariant kernel cannot. A governance-rich protocol provides governance scaffolding by default; an invariant kernel provides none. These are real costs and benefits and the choice between them is substantive. But the choice exists, and pretending it does not is the move both programs make when they treat governance at the substrate layer as constitutive of the kind of arrangement they are analyzing.

Remark 4.1 (On the relationship between governance and adaptability). The most common defense of substrate-layer governance is that adaptability is required: external conditions change, errors need correction, opportunities require parameter updates. This defense is forceful for protocols whose function is itself adaptive (lending markets that must respond to market conditions, scaling layers that must respond to throughput demand, identity systems that must respond to evolving identity standards). It is weaker for protocols whose function is structurally invariant. A protocol that does one well-defined thing (enforce a bonded bilateral commitment) can be invariant in a way that a protocol that does many things subject to external state cannot. The adaptability defense is therefore specific to function, not generic to protocols.

5 The Ungoverned Substrate: A Structural Alternative

We now describe the option that the bonded primitive realizes and that neither contemporary program names.

What "ungoverned" means here. The kernel is governed in no respect that the contemporary governance literature would recognize. There is no proposal process, no voting mechanism, no quorum, no veto, no executive, no foundation that controls protocol evolution, no developer group with privileged update access, no upgrade path of any kind. The deployed bytecode is invariant: identical input produces identical output for as long as the underlying chain operates. To change the kernel's behavior requires deploying a *different kernel* at a different address, which constitutes a different protocol that the existing one has no relationship to.

This is not an absence of governance through inattention. It is governance designed out of the substrate. The kernel's invariants are fixed by mathematics, formal verification, and code, not by the consent of an evolving body of participants. The kernel does not require governance because it does not have decisions to make.

What this requires of the kernel. The ungoverned-substrate option is not free. It requires that the kernel's function be specifiable in advance, that the specification be small enough to be formally verified, and that the resulting bytecode be small enough to be auditable to a standard that warrants trust without ongoing governance. The companion implementation paper documents how this is achieved for the bonded primitive: a kernel of approximately two hundred lines of Solidity, formally verified by multiple independent verification stacks (TLA$^+$, symbolic execution, property-based fuzzing, deductive verification), with a small public surface and no admin functions. The choice to have an ungoverned substrate is purchased by accepting the constraint that the substrate's function be amenable to this discipline.

Most blockchain protocols that have been built do not satisfy the constraint. Lending protocols, scaling solutions, identity systems, prediction markets, and many other categories of on-chain arrangement involve interactions with external state that the protocol must respond to and that no fixed code can anticipate. For those, governance is genuinely required because adaptability is genuinely required. The bonded settlement primitive is in a different category: the primitive's function (enforce a bonded bilateral commitment between consenting parties) is structurally amenable to fixed specification, and that is what makes the ungoverned-substrate option available for it.

The composition admit. A consequence of the ungoverned-substrate design is that the substrate admits arbitrary higher-layer arrangements without constraining them. A community can constitute a network state on top of the substrate, with whatever governance the network state chooses (constitutional, computational-democratic, charismatic, inherited, none). A group of participants can constitute a DAO, with whatever on-chain or off-chain governance the DAO chooses. A cooperative can compose itself, with whatever cooperative governance is appropriate. A peer network can operate without constituting any entity at all. Two strangers can transact once and never again. The substrate accommodates each of these because it makes no commitments about what the parties operating on it should be.

This is the central architectural observation. The substrate's ungoverned character is what makes it neutral with respect to the governance forms that compose on top of it. A governance-rich substrate would import its governance into every arrangement constituted on top of it, because the arrangements would inherit the substrate's governance dependencies. An ungoverned substrate imports nothing; the arrangements that compose on top of it owe the substrate nothing except the use of its public infrastructure, and they are free to govern (or not govern) themselves in whatever way their participants choose.

Remark 5.1 (The substrate is sub-political, not anti-political). The ungoverned substrate is not a rejection of politics or of governance; it is a layer beneath them. The political question of how a network state should be governed, the institutional question of how a DAO should make decisions, the legal question of how an arrangement should integrate with conventional jurisdictions — all of these remain real and unresolved at the layer at which they apply. The substrate neither answers nor displaces them. What the substrate does is make it possible for these questions to be answered *differently in different arrangements* on the same infrastructure, rather than uniformly across all arrangements that share the infrastructure.

6 Where Governance Returns: At the Graph

The argument that the substrate is ungoverned is not the argument that everything constituted on it is ungoverned. To the contrary: most useful arrangements constituted on the substrate will have governance of some kind. The present section identifies where governance returns and what shape it takes.

Assemblies as governed entities. A bonded process tree composed at the substrate is an assembly — the corpus's term for a transaction-scoped arrangement of bonded commitments among multiple parties. A simple assembly (two parties, one commitment) requires no governance: the parties have agreed to the commitment, and the substrate enforces it. A complex assembly (many parties, many commitments, ongoing relationships) typically does require governance: questions arise that the original commitments did not specify, parties join and leave, parameters need to be updated, disputes require resolution beyond what bonding can provide. For complex assemblies, governance is real work that the assembly's participants must do, and the substrate provides no scaffolding for it.

This is a feature, not a gap. The substrate's silence on governance allows assemblies to choose the governance form appropriate to their function. A cooperative assembly can use cooperative governance; a corporate assembly can use corporate governance; a network-state assembly can use whatever constitutional process the network state has settled on; a peer-network assembly can use no formal governance and rely on informal coordination. Each of these is a graph-tier choice that the substrate admits and on which it takes no position.

The COALA scaffolding extends. The legal scaffolding work developed within COALA and adjacent groups remains relevant and useful for assemblies that need it. A DAO constituted on the substrate that wishes to enter into contracts with non-DAO counterparties, hold conventional property, limit liability for participants, or engage with regulators on recognized terms, will benefit from the legal frameworks the COALA program has produced. The scaffolding does not become unnecessary because the substrate is ungoverned; it becomes one option among many for assemblies that elect to participate in conventional legal systems. Other assemblies may elect not to (purely on-chain coordination, transient relationships, agent-only arrangements), and the substrate admits both.

Network states on the substrate. A network-state community constituted on the substrate has its governance work fully in front of it. The substrate provides the missing economic engine the network-state program identified as under-specified, but it provides nothing else: not the constitutional process, not the community self-conception, not the diplomatic framework, not the internal adjudication. These are the network state's to design. What the substrate does is make those designs possible without requiring the network state to also design and govern its own commercial-enforcement infrastructure, which has historically been a substantial fraction of what state-formation involves.

The infrastructural-literacy concern. The blockchain-governance program's concern about infrastructural literacy — that participants generally cannot evaluate the arrangements they are operating under — applies to the substrate as it does to any other infrastructure. The substrate does not reduce the literacy gap; what it does is *not expand* the gap. A governance-rich protocol's literacy demand grows over time as the protocol's rules and parameters evolve; a participant must track the evolution to remain informed. An invariant kernel's literacy demand is bounded: the participant must understand the kernel's fixed rules once and for all. This does not solve the literacy problem (the kernel's rules are non-trivial, and most participants will not in fact read them), but it bounds the problem in a way that governance-rich protocols structurally cannot. The literacy work the blockchain-governance program calls for is necessary; the substrate makes it possible without requiring participants to re-do it on a continuing basis.

Remark 6.1 (On the relationship to confidence-machine analysis). The confidence-machine framing of the blockchain-governance program correctly identifies that trust shifted from human institutions to code does not eliminate trust or governance; it relocates them. The present paper agrees with this for the class of arrangements the framing addresses (governance-rich protocols whose rules evolve). The ungoverned substrate is a different case: trust in the substrate is required only in the substrate's invariant specification, not in any ongoing human governance of it. The trust is therefore mathematizable: formal verification, audit, and the public visibility of the deployed bytecode discharge the trust burden in a way that is not available for governance-rich systems whose evolving parameters cannot be verified in advance. The confidence-machine concern about infrastructural literacy survives; the concern about ongoing-governance accountability does not, because there is no ongoing governance of the substrate to hold accountable.

7 What This Paper Is Not

The argument is liable to several misreadings. We address them explicitly.

Not a critique of the network-state program. The paper does not argue that the network-state program is wrong, that network states are undesirable, or that the program's prescriptions should be rejected. The program's analysis of exit, of community formation, of what it takes to constitute a sovereign-equivalent entity, is substantive on its own terms. The paper's claim is narrower: the substrate that the program needs does not have to be one the network state constitutes and governs. Recognizing the ungoverned-substrate option clarifies what the network-state program is committed to and what is design choice within it; it does not adjudicate whether the program's overall direction is right.

Not a critique of the blockchain-governance program. The paper does not argue that the blockchain-governance program is wrong, that the confidence-machine analysis is incorrect, or that COALA's legal-scaffolding work is misguided. The program's analysis applies to the class of arrangements it was developed for, and it applies correctly. The paper's claim is that the class of governance-rich blockchain arrangements is not the only class; an invariant-kernel architecture is a structurally distinct class to which the confidence-machine concern about ongoing governance does not apply, and to which the COALA scaffolding becomes one option among many for assemblies built on top of the substrate. This is an extension of the program's analytical landscape, not a refutation of it.

Not a claim that all blockchain protocols should be ungoverned. The paper does not argue that governance-rich protocols are mistaken, that adaptability through governance is unnecessary, or that all on-chain coordination should adopt the ungoverned-substrate model. Many protocols have functions that genuinely require adaptability and therefore genuinely require governance: the paper's argument is that the *coordination-substrate function* is not in this class when it is restricted to enforcing bonded bilateral commitments, and that this restriction is what makes the ungoverned-substrate option available for that specific function. Protocols whose function is genuinely adaptive should have governance; the paper does not contest this.

Not a depoliticization of the network question. The paper does not argue that politics is over, that governance has been transcended, or that the substrate change makes political questions irrelevant. Quite the opposite: the substrate change is the precondition under which political questions about network arrangements become legible as graph-tier questions rather than being entangled with substrate-tier governance dependencies. There is more political work to do at the graph layer, not less, because arrangements that previously had to be uniform across the infrastructure can now diverge substantively from one another. The work of designing graph-tier governance for the arrangements that need it is not addressed by the substrate change; it is enabled by it.

Not a constitutional proposal. The paper does not propose any constitution, governance form, or political arrangement for any specific network. The substrate admits arbitrary arrangements; choosing among them is work that the participants of each arrangement must do. The paper's contribution is structural: identifying that the substrate exists as an option, that it is distinct from the governance-at-substrate-layer assumption that contemporary programs share, and that the relocation of governance to the graph is the political-architectural content of recognizing the option.

Not a prediction. Finally, the paper does not predict that the network-state program will adopt the ungoverned-substrate option, that blockchain-governance researchers will modify their analyses to include it, or that any specific arrangement will be built. The argument is conceptual: the option exists, it is distinct from what either program currently presupposes, and recognizing it clarifies the analytical landscape. Whether the recognition occurs and what arrangements emerge from it are empirical questions on which the present paper takes no position.

8 Conclusion

This paper has argued that two contemporary research programs on network-native coordination — the network-state program advanced by Srinivasan [2022] and the blockchain-governance program developed by De Filippi and Wright [2018] and the COALA-affiliated research community — share a structural assumption that the alternative-to-state coordination must itself be a governed network entity. The bonded settlement primitive examined in the corpus's companion mechanism paper realizes a third option that neither program names: an ungoverned substrate, invariant by design, on which arbitrary governed entities can be composed but which itself is not a governed entity at all. Recognizing the option does not adjudicate among governance forms above the substrate; it relocates the governance question from the substrate, where both programs place it, to the graph, where the substrate admits it to be answered in arbitrarily many ways.

The substantive observation is that the choice between a governance-rich substrate and an ungoverned substrate is a real design choice with real consequences. Governance-rich substrates provide adaptability at the cost of ongoing governance-of-the-governance burden; ungoverned substrates provide invariance at the cost of inability to adapt. For substrate functions that are amenable to fixed specification — the bonded settlement primitive being the case the corpus has established — the ungoverned-substrate option is available, and its availability is what permits the diversity of higher-layer governance forms the substrate admits. For substrate functions that are not amenable to fixed specification, the governance-rich option remains the only one available, and the blockchain-governance program's analysis of those arrangements remains directly applicable.

We close with two observations of method, mirroring the closings of the companion papers in the political-philosophy and political-economy clusters. First, the kernel is ideologically agnostic; the graph is the politics — and the present paper extends this formulation in a fourth direction. The companion political-economy papers showed that subordination and coordination rent become graph-tier choices when the substrate removes them as load-bearing variables. The companion political-philosophy paper on coercion showed that coercion becomes a bounded political-philosophical question when the substrate substitutes pre-commitment for retributive enforcement. The present paper shows that governance becomes a graph-tier choice when the substrate is invariant. In each case, the move is the same: a feature that prior architectures treated as constitutive of the substrate becomes a graph-tier choice when the substrate is designed to exclude it. The graph is the politics in a strong sense: it is the layer at which political-architectural choices are made on infrastructure that makes no political-architectural choices itself.

Second, the recursive note. Future architectures may make further substrate variables relocatable to the graph, or may re-introduce variables the present substrate has excluded. The identification of the ungoverned-substrate option is not a terminal architectural achievement; it is the next architectural choice whose conditions and consequences the present moment has made legible. Subsequent moments will face their own choices, on conditions the present paper cannot anticipate. What the substrate change does is enlarge the menu of architectural options visible from the present, not exhaust the architectural options that subsequent moments will have available.

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The argument of this paper began as a proposal to extend the companion political-philosophy paper on coercion to engage two specific contemporary programs by name. The Network State program was already in scope as a remark in that paper; the suggestion to engage Primavera De Filippi and the COALA-aligned research community substantively was the prompt that occasioned the present essay. The COALA collaborators were already acknowledged in the companion political-economy paper on hegemony; the present engagement is meant in the spirit of that acknowledgment, as collaboration with rigorous adjacent work that addresses the same architectural question from a different direction. I thank the interlocutor whose suggestion prompted this paper, and the readers of earlier drafts who helped me calibrate the engagement carefully — the paper is meant neither to absorb the adjacent programs into the corpus's framing nor to dismiss them, but to identify where the corpus's framing supplies an architectural option the adjacent programs have not yet had occasion to name.

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